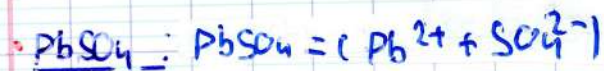


Exercice 7:

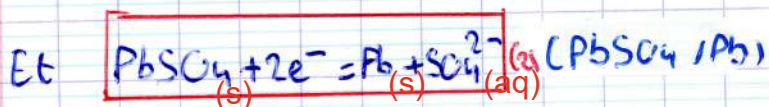
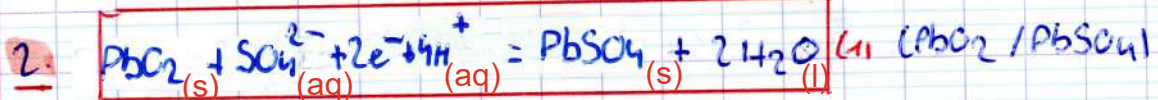
1. Pb: $n.O.(Pb) = 0$



$n.O.(Pb) = 4 - 0 = +IV$



Doc $n.O.(Pb) = +II$



3.
$$E_1 = E_1^0 + \frac{0,06}{2} \log \left(\frac{[SO_4^{2-}]^2 [H^+]^4}{CO_2} \right)$$

A.N.: $E_1 = 1,8 \text{ V}$

$$E_2 = E_2^0 + \frac{0,06}{2} \log \left(\frac{CO}{[SO_4^{2-}]} \right)$$

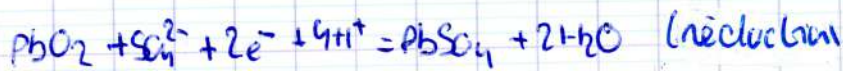
$E_2 = -0,39 \text{ V}$

4. $e = E_1 - E_2$

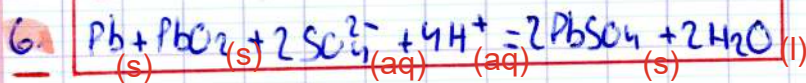
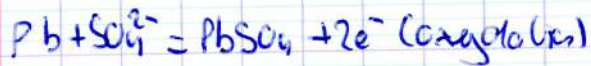
$$e = (E_1^0 - E_2^0) + \frac{0,06}{2} \log \left(\frac{[SO_4^{2-}]^2 [H^+]^4}{CO_2} \right)$$

A.N.: $e = 2,2 \text{ V}$

5. Cathode: (+)



Anode: (-)



loi de Nernst:

$$E_1 = E_1^0 + \frac{0.06}{2} \log \left(\frac{[\text{SO}_4^{2-}]^2 [\text{H}^+]^4}{c^0} \right) \quad \text{et} \quad E_2 = E_2^0 + \frac{0.06}{2} \log \left(\frac{c^0}{[\text{SO}_4^{2-}]} \right)$$

Relation de Gibbs et Wroge: $k^0 = \frac{c^0}{[\text{H}^+]_{\text{eq}}^4 [\text{SO}_4^{2-}]_{\text{eq}}^2}$

Unite à l'eq: $E_1 = E_2$

$$\Rightarrow E_2 - E_1 = (E_2^0 - E_1^0) + \frac{0.06}{2} \log \left(\frac{c^0}{[\text{SO}_4^{2-}]_{\text{eq}}^2 [\text{H}^+]_{\text{eq}}^4} \right) = 0$$

$$\Leftrightarrow \frac{0.06}{2} \log(k) = E_1^0 - E_2^0$$

$$\Rightarrow k = 10^{\frac{2}{0.06} (E_1^0 - E_2^0)}$$

A.N: $k = 4,6 \times 10^{68}$

Réactif limitant: PbO_2
 $k_{\text{simax}} = n(\text{PbO}_2)_{\text{initial}}$

7. $Q = n \int_{\text{max}} \vec{J}$ avec $\int_{\text{max}} = \frac{m(\text{PbO}_2)}{239} = 0,50$ et $n=2$

A.N: $Q = 9,69 \times 10^4 \text{ C}$ donc $Q = 26,9 \text{ A.h}$